

Report writing

Stage 3

Reports

- Reports in the Advanced Lab should be **short** and concise
- The purpose of your report is to communicate your results in a clear and concise way such that the experiment could be repeated by the reader and results compared.
- Your understanding will be assessed in the discussions you have with the teaching staff.
- The style of the report should be like a journal article e.g. the style used in the *American Journal of Physics*

The anatomy of a lab report

- The laboratory report is normally divided into a series of distinct sections: *Abstract, Introduction, Theory, Apparatus and Experimental Procedure, Results, Analysis, Discussion, Conclusions, References and Appendices*

The abstract

A report normally opens with a paragraph summarising the purpose of the experiment, the main findings and the conclusions reached.

The introduction

The introduction should describe the aims of the experiment and provide the context or motivation for the experiment.

Theory section

A **short** outline of the theory necessary for understanding the results should be presented including appropriate references. It is not necessary to derive equations but rather present a summary of the key equations.

Cite your references either by number (i.e., [1], [2], etc...) or by author and year [e.g., Shanker et al (1985), George(1989), ...]. A list of references used must be included at the end of the report.

Experimental procedure section

This section should describe what was actually done.

It should provide enough information on the experimental arrangement, measuring conditions and procedures taken so that the reader could use the description to replicate the experiment.

Diagrams are essential(photographs are not necessarily useful)

Useful or not?

3. Methodology



Fig 3.1 Photograph of Apparatus

A photograph of the experimental set up is provided in fig 3.1. The circuit diagram for the experiment is provided in fig 3.2. A long solenoid consisting of $N_1 = 1013$ turns of

Critique this:

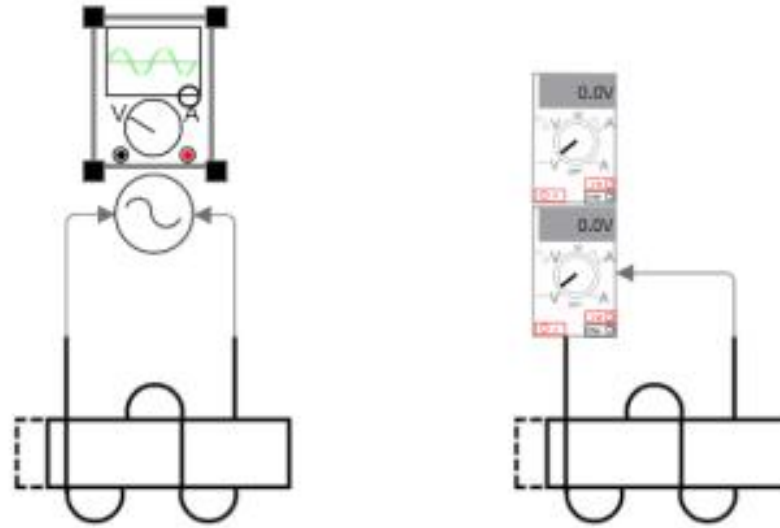


Figure 4: Diagram of circuit used in experiment

Results

- Report relevant data, observations and findings arising from your experiment should be presented in a clear, logical sequence.
- Although some raw data can be included in the report avoid presenting large blocks of unprocessed data. Instead, when possible, try to summarise the data in the form of tables or graphs.



Which of the following are acceptable?...

Exhibit A

ν (brass) in Hz	V in volts	$V\nu^{-1}$	$\nu^{-1/2}$
5000	7.72E-03	1.54E-06	0.014142
10000	1.30E-02	1.3E-06	0.01
15000	1.73E-02	1.15E-06	0.008165
20000	2.11E-02	1.05E-06	0.007071
25000	2.56E-02	1.03E-06	0.006325
30000	2.98E-02	9.94E-07	0.005774
35000	3.45E-02	9.86E-07	0.005345
40000	3.86E-02	9.65E-07	0.005
45000	4.33E-02	9.63E-07	0.004714
50000	4.99E-02	9.97E-07	0.004472

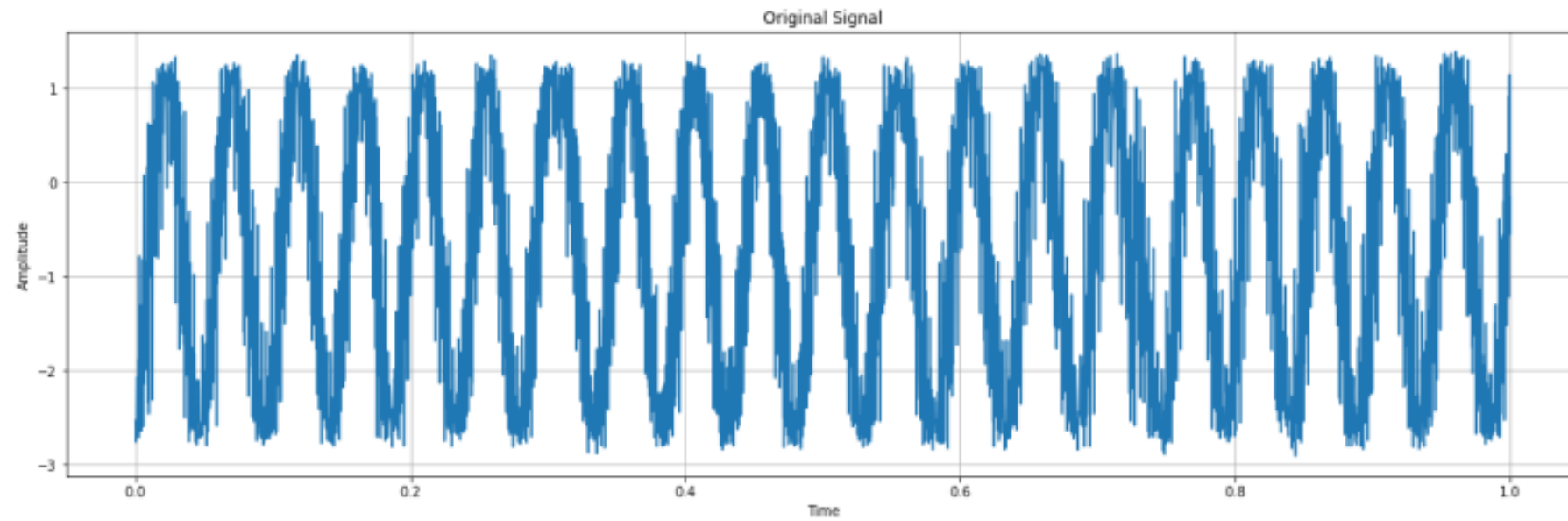
Fig 4.1 Experimental values obtained for brass sample

Exhibit B

zS

1945.3347168	627.6709595	627.6709595	0.0000000	-663.2834473	-13.3547010
627.6709595	1945.3347168	627.6709595	0.0000000	13.3547010	592.0584106
627.6709595	627.6709595	1945.3347168	0.0000000	13.3547010	574.2521362
0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
-663.2834473	13.3547010	13.3547010	0.0000000	1945.3347168	-627.6709595
-13.3547010	592.0584106	574.2521362	0.0000000	-627.6709595	1945.3347168
-13.3547010	574.2521362	592.0584106	0.0000000	-627.6709595	627.6709595
0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
-912.5712280	-627.6709595	-253.7393188	0.0000000	592.0584106	13.3547010
-627.6709595	-912.5712280	-253.7393188	0.0000000	-13.3547010	-663.2834473
253.7393188	253.7393188	235.9330444	0.0000000	-574.2521362	-13.3547010
0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
592.0584106	-13.3547010	574.2521362	0.0000000	-912.5712280	627.6709595
13.3547010	-663.2834473	13.3547010	0.0000000	627.6709595	-912.5712280
574.2521362	-13.3547010	592.0584106	0.0000000	-253.7393188	253.7393188
0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
592.0584106	574.2521362	-13.3547010	0.0000000	-912.5712280	253.7393188
574.2521362	592.0584106	-13.3547010	0.0000000	-253.7393188	235.9330444
13.3547010	13.3547010	-663.2834473	0.0000000	627.6709595	-253.7393188
0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
-912.5712280	-253.7393188	-627.6709595	0.0000000	592.0584106	-574.2521362
253.7393188	235.9330444	253.7393188	0.0000000	-574.2521362	592.0584106
-627.6709595	-253.7393188	-912.5712280	0.0000000	-13.3547010	13.3547010

Exhibit C



Graphs

Use error bars

Use *curve_fit*

Report on the P
value or goodness
of fit

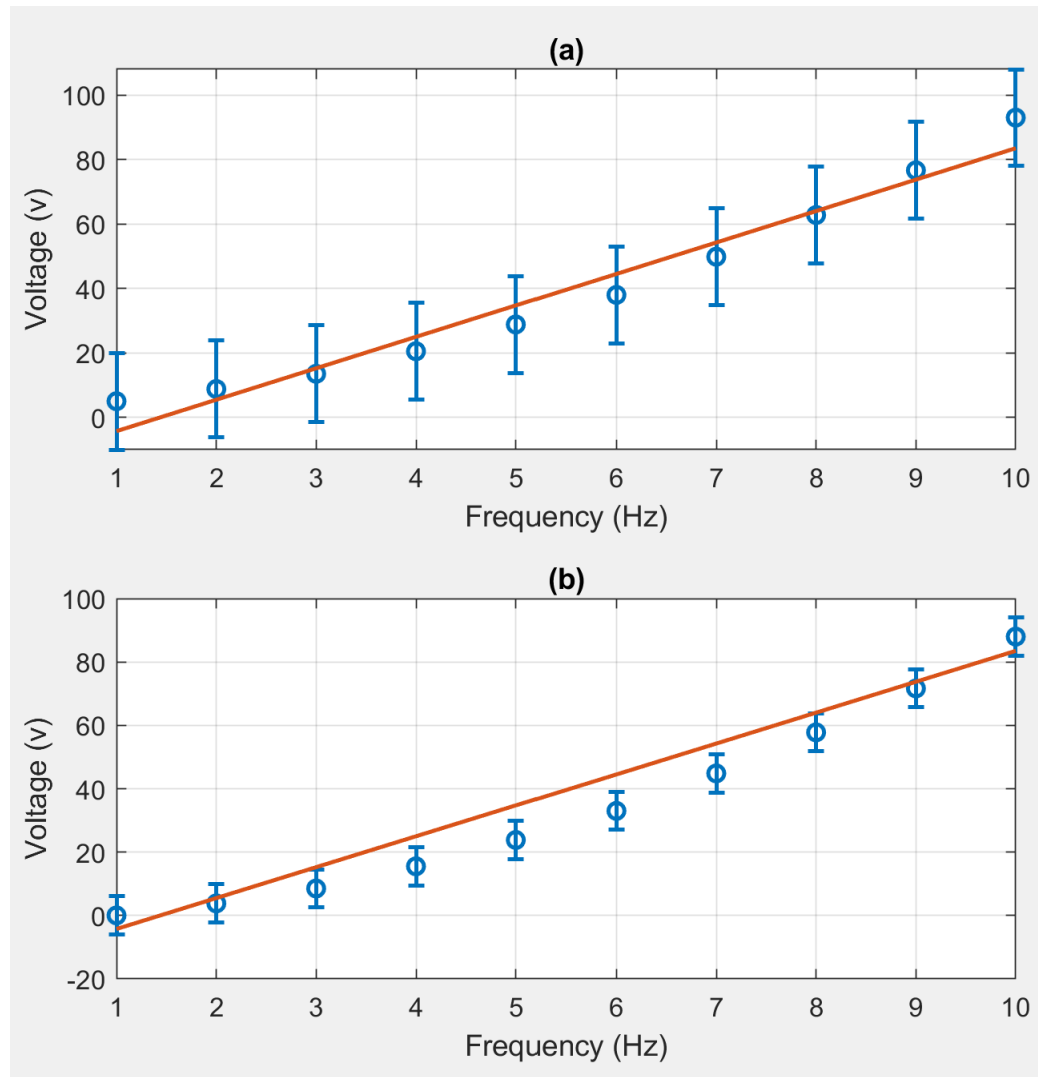


Figure 2. Plots of the Voltage measured in volts versus frequency in Hertz. The voltage was measured across the primary coil and the frequency was applied to the secondary coil

Discussion + conclusion

- Discuss the results in relation to the theory presented in the introduction.
- Are your results in good agreement with the theory?
- What have you learned from the measurements?
- Why did the experiment or a particular part of it fail?
- Have you seen anything unexpected?
- What are the weaknesses and strengths of the experimental approach employed and how could the experiment be improved?

Remember we are not necessarily looking for the correct answer but rather your ability to interpret your results correctly and analyse the strengths and weaknesses of your data.

Appendix

- At the end of the report you must include a copy of any Python codes or Jupyter Notebook pages that you used in treating the raw data, or code written to interface with various bits of equipment.
- These can be converted to pdf and imported into the report.
- Your code should be well documented in order to make it easy to interpret

References and Citations

References (books, articles etc) used in the experiments or in the preparation of the report should be listed at the end

References and Citations are required in Academic Writing to:

- respect and acknowledge the thoughts, ideas and creations from other people's work
- give credibility to the arguments and conclusions in an assignment
- provide evidence for the ideas and arguments made in an assignment
- avoid plagiarising or representing another person's work as your own
- let others trace the sources and materials used to create your work*

*IMPORTANT NOTE: By including a reference list with information such as Book Title, Publisher, Publication Date, Journal Title, Website address etc., you are allowing your readers examine the evidence used in your assignment and read more about the topic.

Referencing styles

- Style guides are available from the library website
- <https://libguides.ucd.ie/academicintegrity>
- I prefer the following based on the American Physics Society style guide

In 1916, the year after the final formulation of the field equations of general relativity, Albert Einstein predicted the existence of gravitational waves. He found that the linearized weak-field equations had wave solutions: transverse waves of spatial strain that travel at the speed of light, generated by time variations of the mass quadrupole moment of the source [1,2]. Einstein understood that gravitational-wave amplitudes would be remarkably

Each reference is given a number in square brackets e.

At the end of the main text there should be a list of the corresponding references

References

- [1] A. Einstein, Sitzungsber. K. Preuss. Akad. Wiss. **1**, 688 (1916).
- [2] A. Einstein, Sitzungsber. K. Preuss. Akad. Wiss. **1**, 154 (1918).
- [3] P. R. Saulson, *Gen. Relativ. Gravit.* **43**, 3289 (2011).
- [4] K. Schwarzschild, Sitzungsber. K. Preuss. Akad. Wiss. **1**, 189 (1916).
- [5] D. Finkelstein, *Phys. Rev.* **110**, 965 (1958).